

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
getwd() #check working directory
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#load packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(trend)
library(Kendall)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
# Set theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1 import data from Ozone_TimeSeries folder
EPA2010 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"), stringsAsFactors=FALSE)
EPA2011 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"), stringsAsFactors=FALSE)
EPA2012 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"), stringsAsFactors=FALSE)
EPA2013 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"), stringsAsFactors=FALSE)
EPA2014 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"), stringsAsFactors=FALSE)
EPA2015 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"), stringsAsFactors=FALSE)
EPA2016 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"), stringsAsFactors=FALSE)
EPA2017 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"), stringsAsFactors=FALSE)
EPA2018 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"), stringsAsFactors=FALSE)
EPA2019 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"), stringsAsFactors=FALSE)

#combine into 1 df
GaringerOzone<- rbind(EPA2010,EPA2011,EPA2012,EPA2013,EPA2014,EPA2015,EPA2016,EPA2017,EPA2018,EPA2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3 set class as data
GaringerOzone$Date <- mdy(GaringerOzone$Date)

# 4 select 3 columns
GaringerOzone.processed <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5 create daily dataset
Days <- as.data.frame(seq(from = as.Date("2010-01-01"),
                          to = as.Date("2019-12-31"),
                          by="day"))
colnames(Days) <- "Date" #rename column name

# 6 combine the data frames
GaringerOzone <- left_join(Days, GaringerOzone.processed) #left join returns all rows from Days, and NU
```

```
## Joining with 'by = join_by(Date)'
```

```
summary(GaringerOzone) #63 days w/ NAs
```

```
##      Date      Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## Min.   :2010-01-01 Min.   :0.00200 Min.   : 2.00
## 1st Qu.:2012-07-01 1st Qu.:0.03200 1st Qu.: 30.00
## Median :2014-12-31 Median :0.04100 Median : 38.00
## Mean   :2014-12-31 Mean   :0.04163 Mean   : 41.57
## 3rd Qu.:2017-07-01 3rd Qu.:0.05100 3rd Qu.: 47.00
## Max.   :2019-12-31 Max.   :0.09300 Max.   :169.00
##      NA's      :63      NA's      :63
```

Visualize

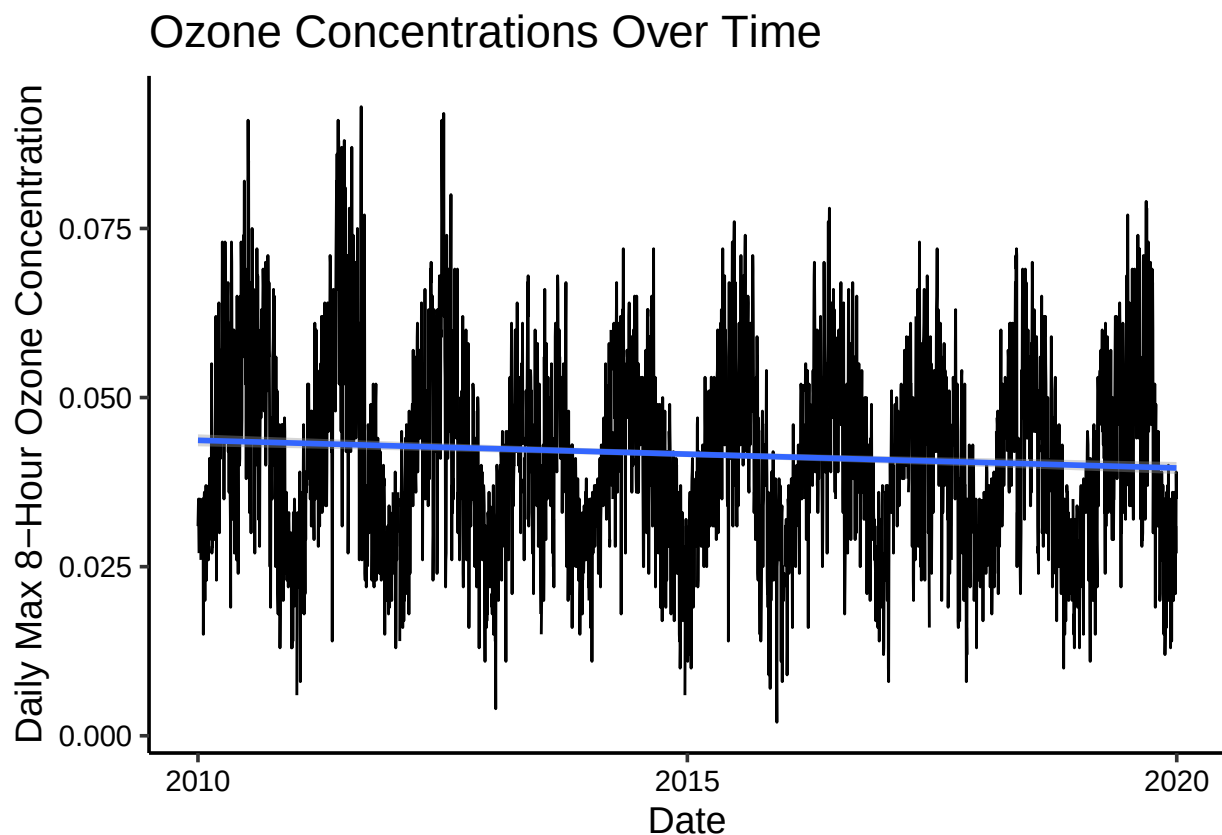
7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7 plot ozone over time
ggplot(GaringerOzone, aes(x=Date, y=Daily.Max.8.hour.Ozone.Concentration))+
  geom_line() +
  geom_smooth(method="lm") +
  labs(title = "Ozone Concentrations Over Time",
       y = "Daily Max 8-Hour Ozone Concentration")+
  theme(mytheme)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```



Answer: Yes, the linear regression suggests a weak negative trend in ozone concentration over time. There also appears to be a strong seasonal component

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8 fill in missing data w/ interpolation
summary(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration) #63 NAs
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## 0.00200 0.03200 0.04100 0.04163 0.05100 0.09300      63
```

```
GaringerOzone_clean <-
  GaringerOzone %>%
  mutate( Daily_Ozone_clean = zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration) ) #create new column

summary(GaringerOzone_clean$Daily_Ozone_clean) #no NAs
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer: We used liner interpolation, which fills in missing data by drawing a line between the nearest two points. We have small gaps in the data, so this method can be used to estimate between points.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9 new df for mean ozone concentrations for each month.
GaringerOzone.monthly <- GaringerOzone_clean %>%
  mutate(GaringerOzone_clean,
    Year = year(Date),
    Month = month(Date)) %>% #add columns for Year and Month
  group_by(Year, Month) %>% #group by year/month
  summarize(
    mean_Ozone = mean(Daily_Ozone_clean) #summarize mean
  )
```

```
## 'summarise()' has grouped output by 'Year'. You can override using the
## '.groups' argument.
```

```
#new Date column with each month-year combination
GaringerOzone.monthly.graph <- GaringerOzone.monthly %>%
  mutate(Date = as.Date(paste0(Year, "-", Month, "-01"))
  )
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10 generate ts objects

## Daily
```

```

#specify dates/freq
f_month <- month(first(GaringerOzone_clean$Date)) #1
f_year <- year(first(GaringerOzone_clean$Date)) #2010

GaringerOzone.daily.ts <- ts(GaringerOzone_clean$Daily_Ozone_clean,
                             frequency=365, start=c(f_month, f_year))

## Monthly
#specify dates/freq
first_month <- month(first(GaringerOzone.monthly.graph$Date)) #1
first_year <- year(first(GaringerOzone.monthly.graph$Date)) #2010

GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly.graph$mean_Ozone
                               ,frequency=12, start=c(first_year, first_month))

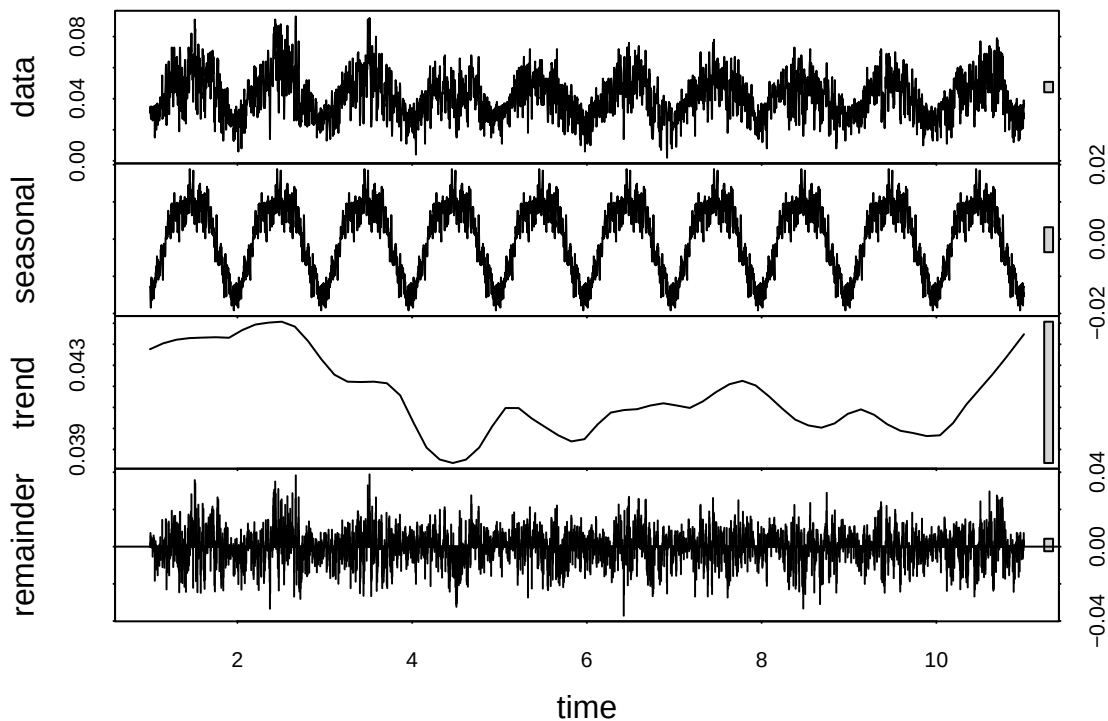
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

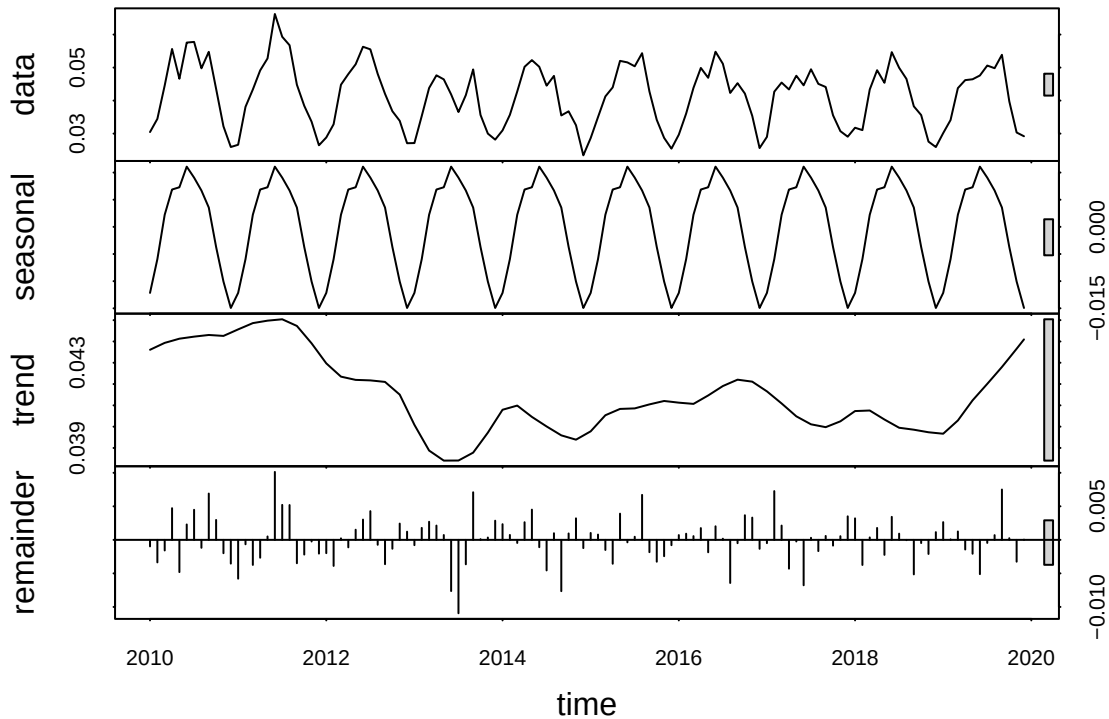
```

#11 decompose ts
##daily
GaringerOzone.daily_Decomposed <- stl(GaringerOzone.daily.ts, s.window = "periodic")
plot(GaringerOzone.daily_Decomposed)

```



```
##monthly
GaringerOzone.monthly_Decomposed <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(GaringerOzone.monthly_Decomposed)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

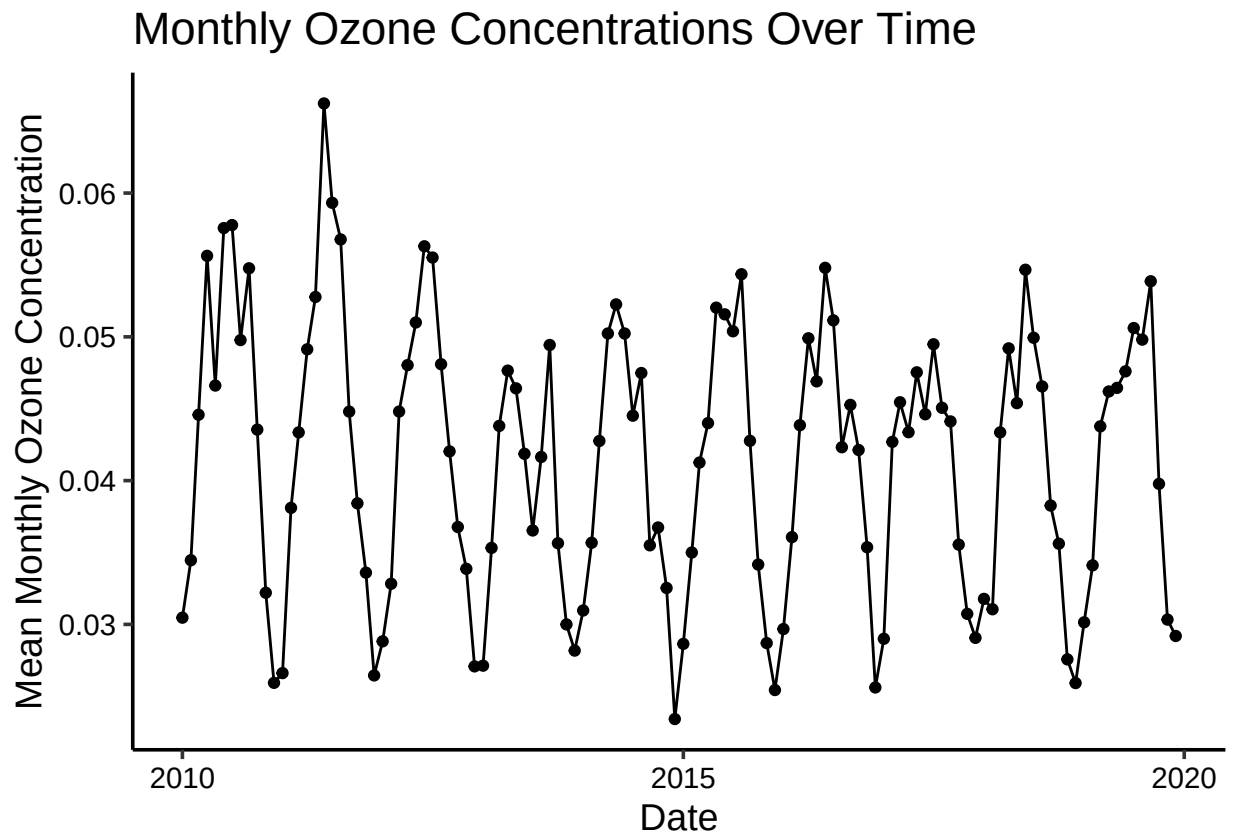
```
#12 monotonic trend analysis - Seasonal Mann-Kendall
##monthly ozone series
monthly_trend <- trend::smk.test(GaringerOzone.monthly.ts)
monthly_trend #p-value = 0.04965, true S is not equal to 0
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
## S varS
## -77 1499
```

Answer: The Seasonal Mann-Kendal test is the most appropriate for seasonal data. Ozone concentrations vary with season, being higher in the summer and lower in the winter.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly.graph,
  aes(x=Date,y=mean_Ozone)) +
  geom_point()+
  geom_line()+
  labs(title = "Monthly Ozone Concentrations Over Time",
    y="Mean Monthly Ozone Concentration")+
  theme(mytheme)
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: We are looking at whether ozone concentrations have changed at this station over the 2010s. The Seasonal Mann-Kendall trend test indicates a slightly negative trend ($z = -1.963$, $p\text{-value} = 0.04965$). The graph reflects a strong seasonal component to ozone.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.


```
#15 Subtract the seasonal component from monthly
```

```
non_seasonal <- GaringerOzone.monthly.ts - GaringerOzone.monthly_Decomposed$time.series[,1]
non_seasonal
```

```
##           Jan           Feb           Mar           Apr           May           Jun
## 2010 0.04263190 0.04041003 0.04234881 0.04875492 0.03932081 0.04647348
## 2011 0.03877706 0.04405289 0.04112300 0.04225492 0.04548211 0.05514015
## 2012 0.04098674 0.03877333 0.04257462 0.04115492 0.04370791 0.04520681
## 2013 0.03929319 0.04126717 0.04157462 0.04077159 0.03912727 0.03077348
## 2014 0.04313190 0.04162432 0.04052623 0.04335492 0.04496598 0.03914015
## 2015 0.04080932 0.04094575 0.03902623 0.03712159 0.04474017 0.04047348
## 2016 0.04184158 0.04201471 0.04162300 0.04302159 0.03961114 0.04370681
## 2017 0.04116416 0.04864217 0.04321978 0.03648826 0.04024017 0.03352348
## 2018 0.04393835 0.03699932 0.04112300 0.04232159 0.03809501 0.04357348
## 2019 0.04230932 0.04005289 0.04154236 0.03932159 0.03915952 0.03650681
##           Jul           Aug           Sep           Oct           Nov           Dec
## 2010 0.04871023 0.04307797 0.05120815 0.04725211 0.04226527 0.04087082
## 2011 0.05025862 0.05007797 0.04124148 0.04212308 0.04366527 0.04138695
## 2012 0.04645216 0.04140056 0.03847481 0.04047792 0.04393193 0.04201598
## 2013 0.02746829 0.03494894 0.04587481 0.03934888 0.04006527 0.04311275
## 2014 0.03545216 0.04078765 0.03194148 0.04044566 0.04259860 0.03835469
## 2015 0.04132313 0.04765862 0.03920815 0.03786501 0.03876527 0.04037082
## 2016 0.04208120 0.03562636 0.04170815 0.04583275 0.04543193 0.04054824
## 2017 0.04041991 0.03836830 0.04055815 0.03925211 0.04079860 0.04399985
## 2018 0.04087152 0.03985217 0.03470815 0.03931662 0.03763193 0.04085469
## 2019 0.04154894 0.04311023 0.05030815 0.04347792 0.04039860 0.04412888
```

```
#16 Mann Kendall test
```

```
non_seasonal_trend <- trend::mk.test(non_seasonal)
non_seasonal_trend #p-value = 0.00754, true S is not equal to 0
```

```
##
## Mann-Kendall trend test
##
## data: non_seasonal
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##           S           varS           tau
## -1.179000e+03 1.943657e+05 -1.651376e-01
```

Answer: After removing the seasonal component, we can run a Mann-Kendall trend test. There is a negative trend, meaning here is a decrease in ozone over time ($z = -2.672$, $p\text{-value} = 0.00754$). The trend is stronger after removing the seasonal component of ozone.